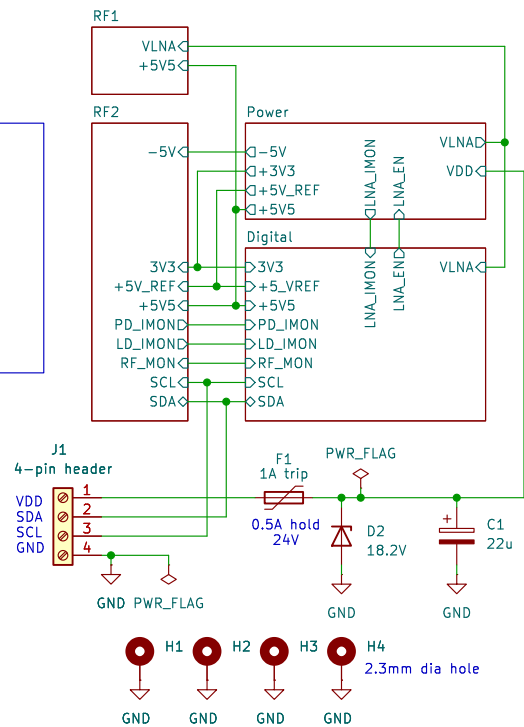


I2C Addresses:

Digipot – 0x2C
Attenuator – 0x20
ADC – 0x10
TEMP & ID – 0x48



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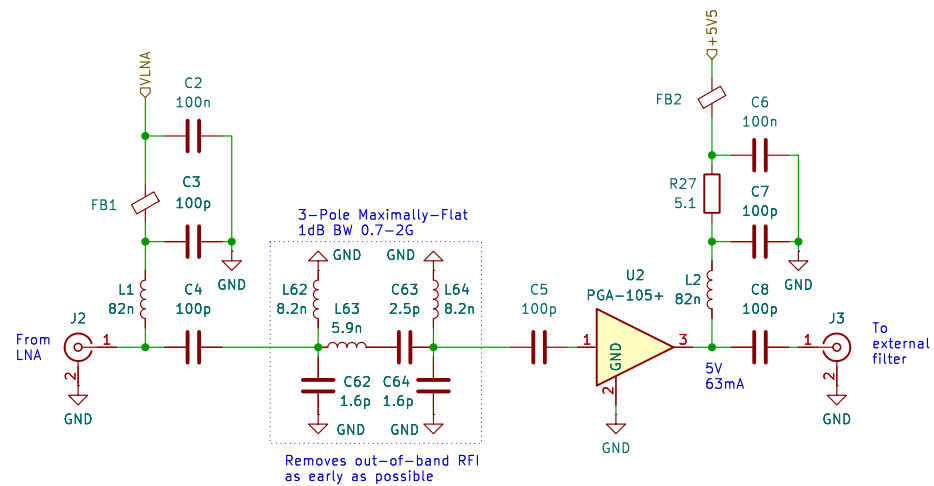

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Title: FTX schematic

Size: A4 Date: 2024-07-01
KiCad E.D.A. 8.0.3

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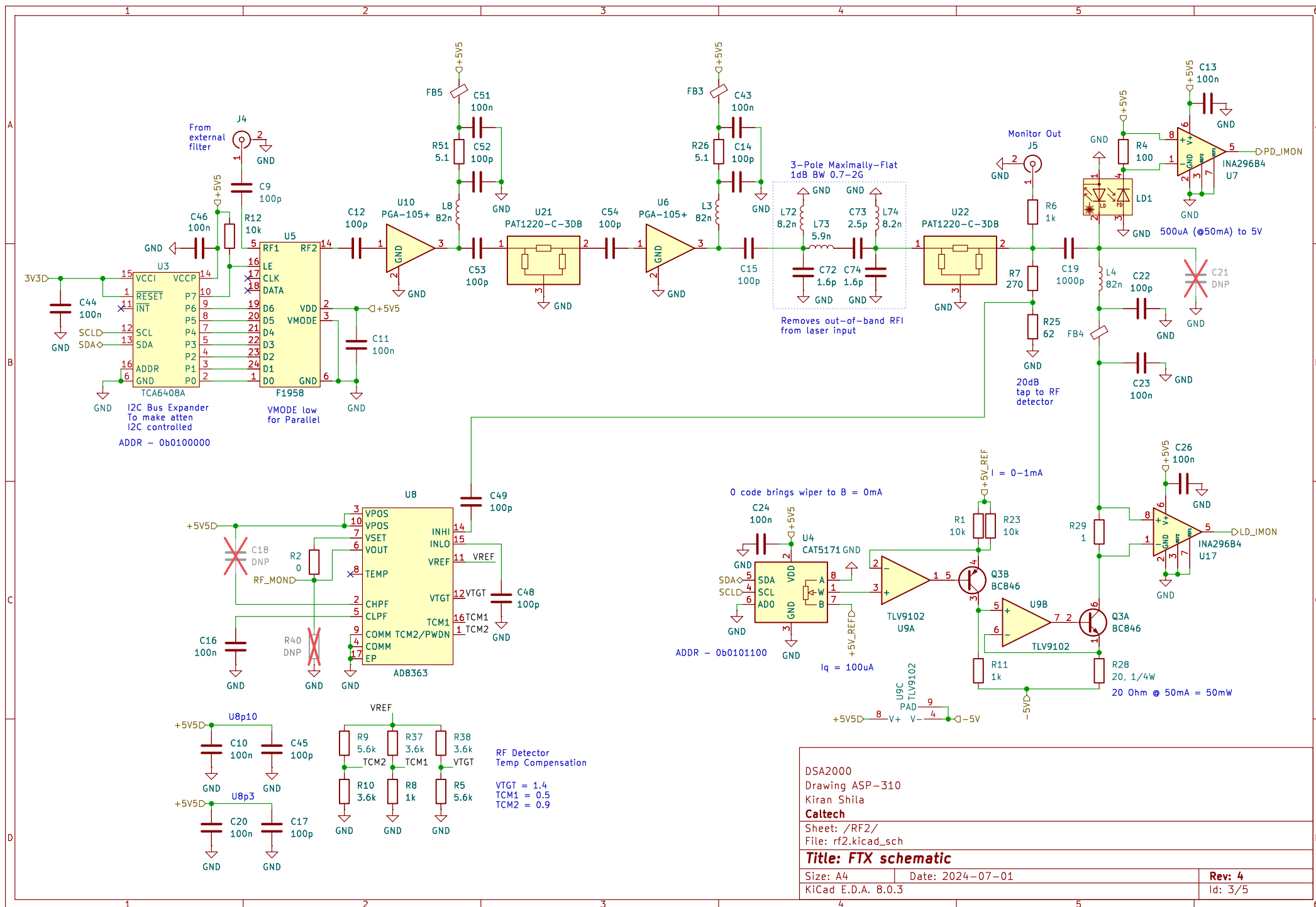
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Size: A4 Date: 2024-07-01

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Rev: 4

Id: 2/5



ADC & digital I/O

Temp Sensor & ID

Weak I2C Pullups
Technically, the pullups should be on the controller,
so these are weak to support that case.

up to VDD=18V
so divide by 4 to keep
ADC input below 5V

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Sheet: /Digital/
File: digital.kicad_sch
Title: FTX schematic
Size: A4
Date: 2024-07-01
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Rev: 4
Id: 4/5

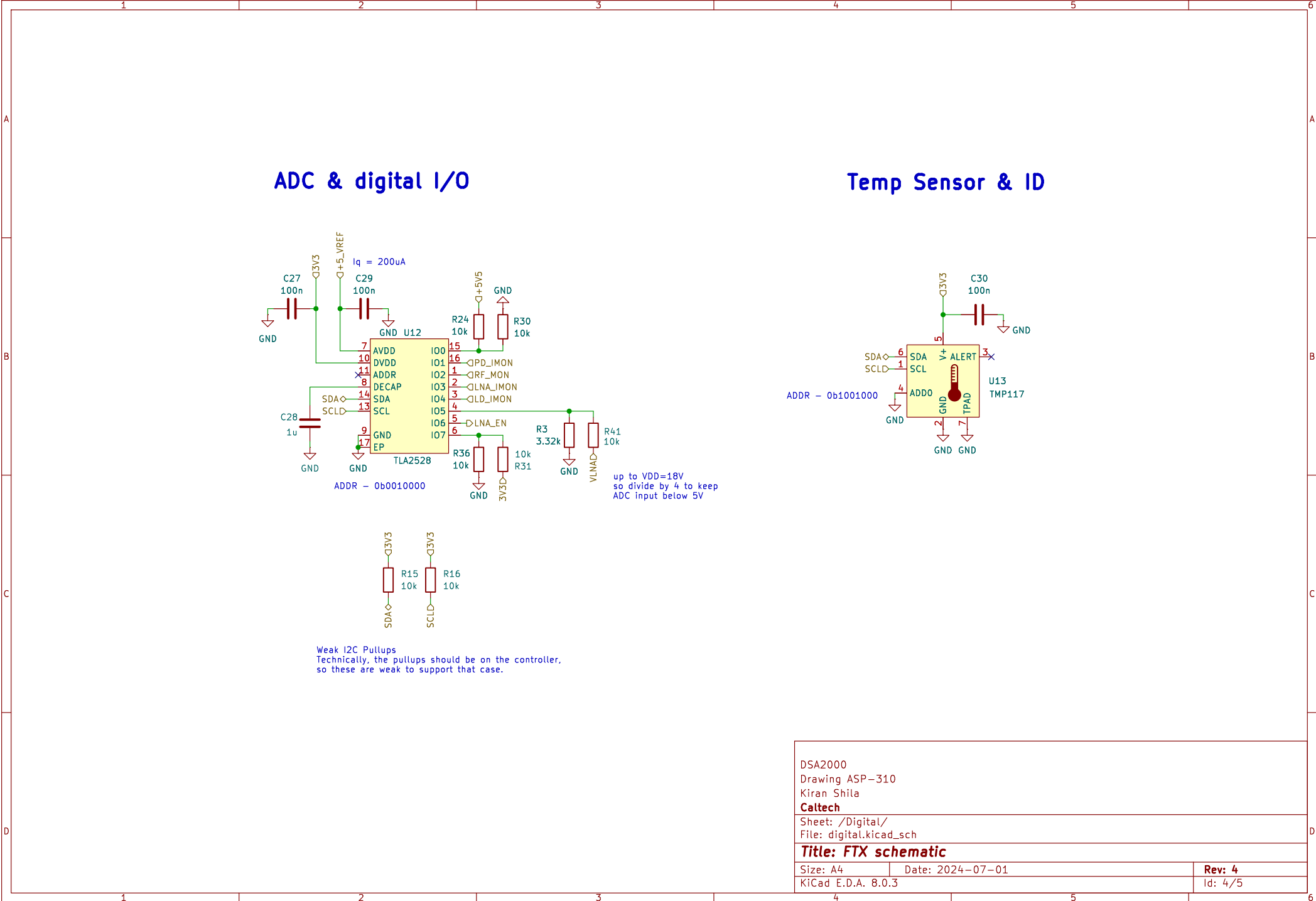
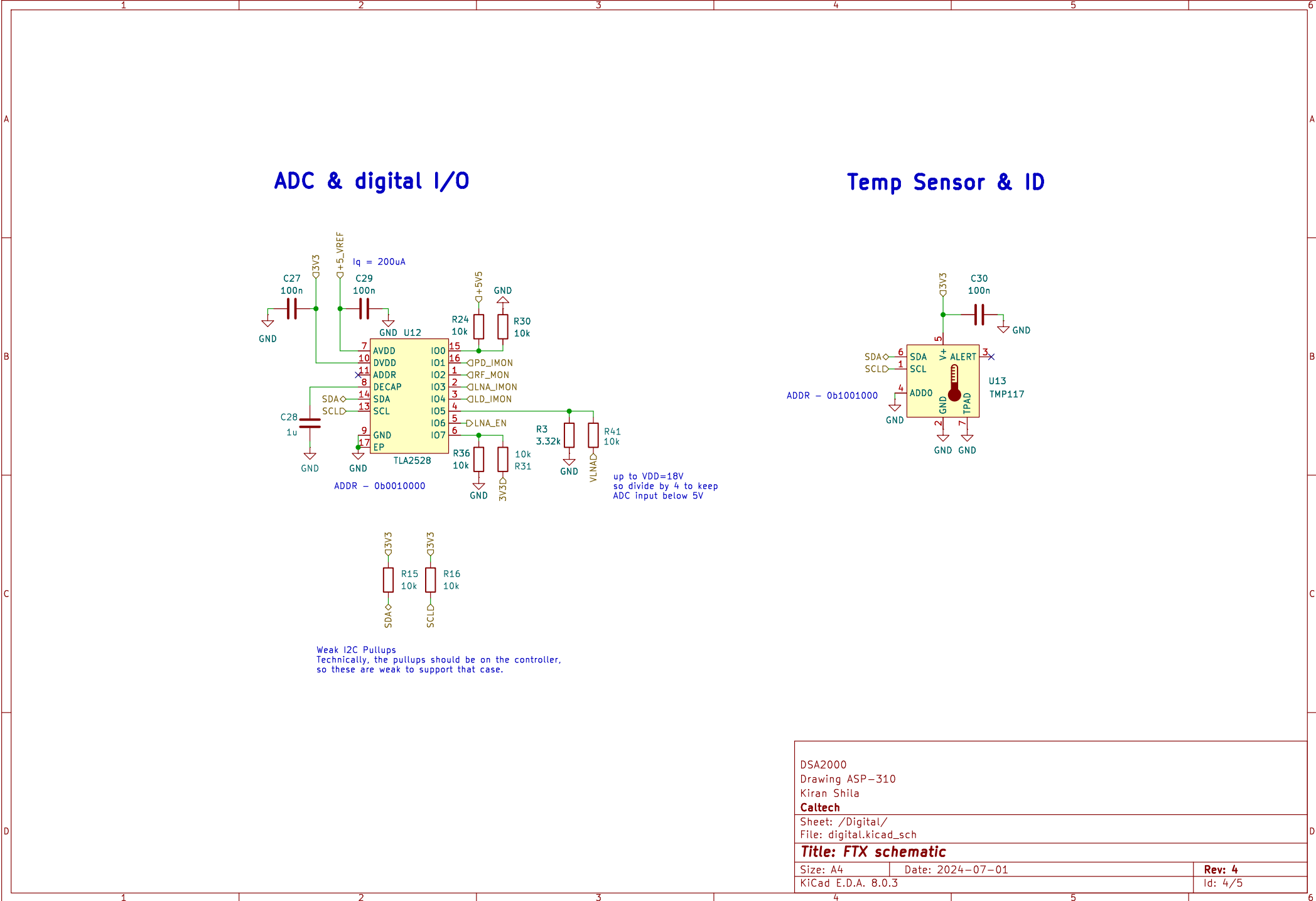
ADC & digital I/O

Temp Sensor & ID

Weak I2C Pullups
Technically, the pullups should be on the controller,
so these are weak to support that case.

up to VDD=18V
so divide by 4 to keep
ADC input below 5V

DSA2000
Drawing ASP-310
Kiran Shila
Caltech
Sheet: /Digital/
File: digital.kicad_sch
Title: FTX schematic
Size: A4
Date: 2024-07-01
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Rev: 4
Id: 4/5



ADC & digital I/O

Temp Sensor & ID

Weak I2C Pullups
Technically, the pullups should be on the controller,
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up to VDD=18V
so divide by 4 to keep
ADC input below 5V

DSA2000
Drawing ASP-310
Kiran Shila
Caltech
Sheet: /Digital/
File: digital.kicad_sch
Title: FTX schematic
Size: A4
Date: 2024-07-01
KiCad E.D.A. 8.0.3
Rev: 4
Id: 4/5

ADC & digital I/O

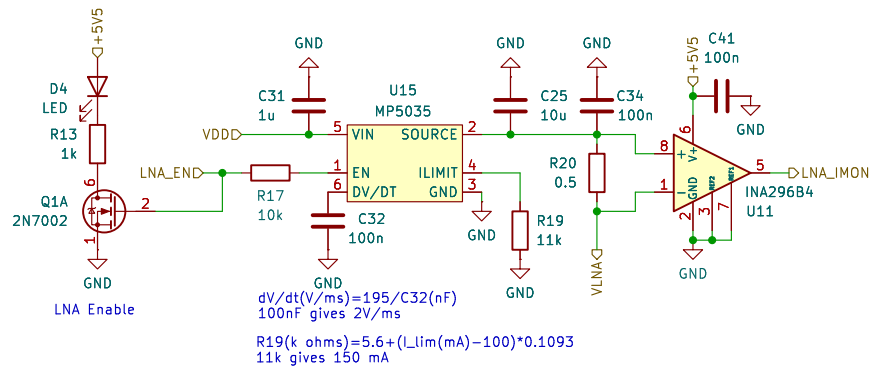
Temp Sensor & ID

Weak I2C Pullups
Technically, the pullups should be on the controller,
so these are weak to support that case.

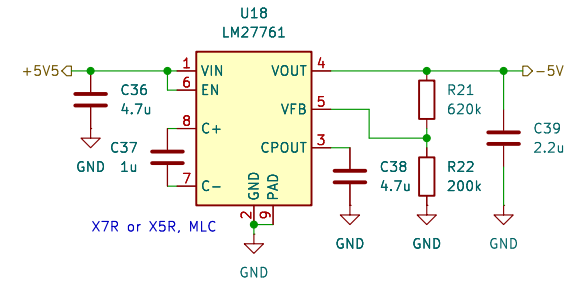
up to VDD=18V
so divide by 4 to keep
ADC input below 5V

DSA2000
Drawing ASP-310
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Caltech
Sheet: /Digital/
File: digital.kicad_sch
Title: FTX schematic
Size: A4
Date: 2024-07-01
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Id: 4/5

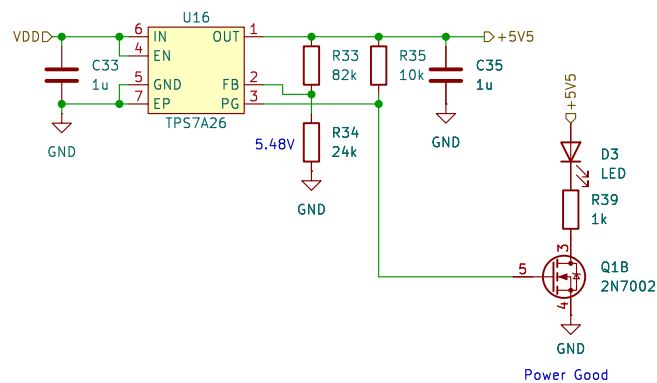
LNA Bias Control



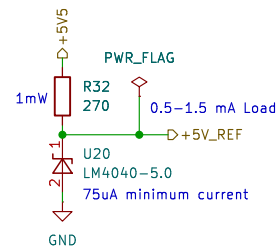
-5V Laser Bias



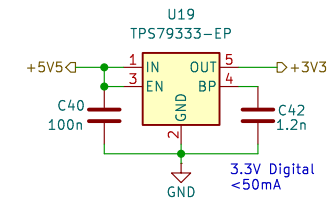
+5.5V Regulated Supply



+5V Precision Reference



+3.3V Digital Supply



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Sheet: /Power/
File: power.kicad_sch

Title: FTX schematic

Size: A4	Date: 2024-07-01
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KiCad E.D.A. 8.0.3

Rev: 4

Id: 5/5